

Gravitational Force:

It is given by Newton's law of gravitation

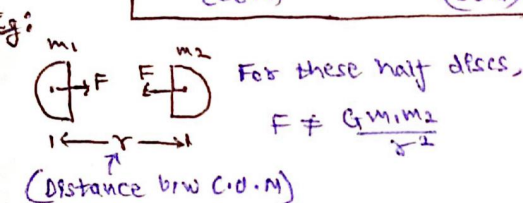
$$F = G \frac{m_1 m_2}{r^2} \quad (G = 6.67 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2})$$

$$\vec{F}_{21} = \frac{G m_1 m_2}{|\vec{r}_2 - \vec{r}_1|} (\vec{r}_2 - \vec{r}_1)$$



→ This law is valid only for point masses or spherical bodies whose mass can be considered to be conc. at centre.

→ Also, Centre of Mass $\neq$ Centre of Gravity (COG)

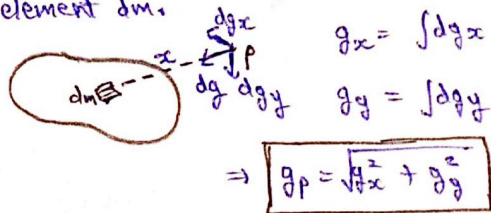


Gravitational Field Intensity:

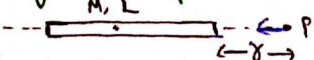
→ It is always directed towards the mass.

For a point mass, $g = \frac{Gm}{x^2}$

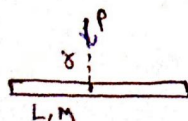
→ For any random body, take a differential element dm .



1) g due to Uniform Rod ⇒



Here, C.O.G would lie at right half of rod because g at P is more due to right half as it is nearer.



$$g = \frac{2GM}{x\sqrt{x^2 + L^2}}$$

2) g due to Ring ⇒

→ g due to complete ring at centre is zero.



$$F_{net} = 0 \Rightarrow g_{net} = 0$$

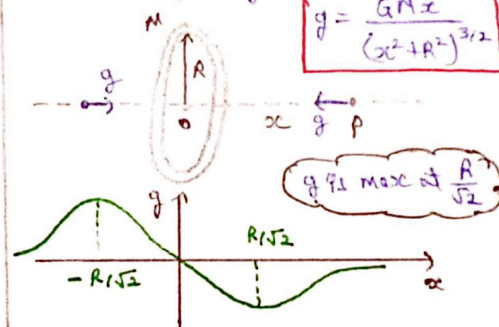
→ If a small part of width w is removed from ring,



$$g = \frac{Gm}{R^2}$$

$$(m = \frac{M}{2\pi R} \cdot w)$$

→ At axis of ring



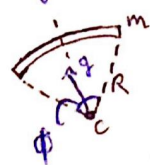
$$g = \frac{GMx}{(x^2 + R^2)^{3/2}}$$

(g is max at $\frac{R}{\sqrt{2}}$)

(Taking radially inward as -ve dir'n)

3) g due to Circular Arc ⇒

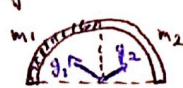
→ g at centre is always along the angle bisector of the arc.



$$g = \frac{2Gm \sin(\phi/2)}{\phi R^2}$$

(Put φ in radians)

Eg: Arc made by two diff materials.

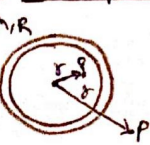


$$g = \sqrt{g_1^2 + g_2^2}$$

Take $\phi = \frac{\pi}{2}$ for both

4) g due to sphere ⇒

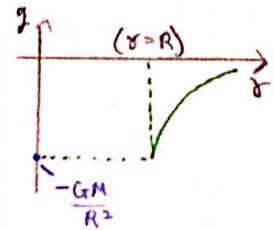
1) Hollow spherical shell



$$g_{in} = 0$$

$$g_{out} = \frac{GM}{r^2}$$

(Same as point mass)



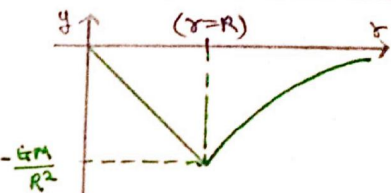
(Taking radially inward as -ve direction)

2) Solid sphere

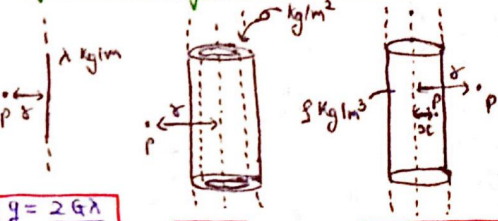


$$g_{in} = \frac{GM}{R^3} r$$

$$g_{out} = \frac{GM}{r^2}$$



3) g due to long wire/cylinder ⇒



$$g = \frac{2G\lambda}{r}$$

$$g_{in} = 0$$

$$g_{out} = \frac{2G\lambda}{r}$$

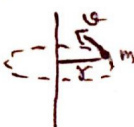
$$g_{out} = \frac{2G\lambda}{r}$$

$$g_{in} = \frac{2G\lambda}{r}$$

$$(\lambda = \sigma \times 2\pi R)$$

$$(\lambda = \rho \times \pi R^2; \lambda m = \rho \times \pi R^2)$$

For a particle revolving around the long wire or cylinder, velocity is independent of its distance.

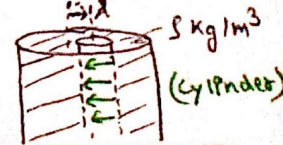
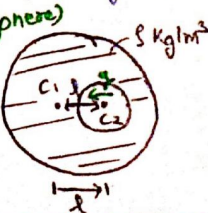


$$F_g = mg \Rightarrow \frac{2G\lambda m}{r} = \frac{mv^2}{r}$$

$$v = \sqrt{2G\lambda}$$

6) g inside cavity ⇒

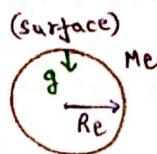
→ g inside cavity is always uniform and always opp. to the vector joining both the centres.



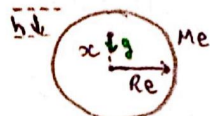
$$\vec{g}_{cav} = -\frac{4}{3} G \rho \pi \vec{r}$$

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g due to Earth:



(h distance inside)



$$g = \frac{GM_e}{R_e^2}$$

$$g_{in} = \frac{GM_e x}{R_e^3} = \frac{GM_e}{R_e^2} \left(\frac{R_e - h}{R_e} \right)$$

(h distance above)

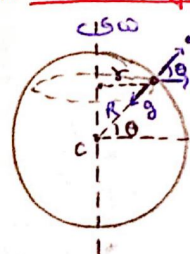
$$g_{out} = g \left(1 - \frac{h}{R_e} \right)$$



$$g_{out} = \frac{GM_e}{(R_e + h)^2} \quad (\text{when } h \text{ \& } R_e \text{ are comparable})$$

$$g_{out} = g \left(1 - \frac{2h}{R_e} \right) \quad (\text{when } h \ll R_e)$$

Variation of g with Earth's Rotation:



$$g_{eff} = g - \omega^2 R \cos \theta$$

$$\Rightarrow g_{eff} = g - \omega^2 R \cos^2 \theta$$

For earth

$$\omega = \frac{2\pi}{86400}$$

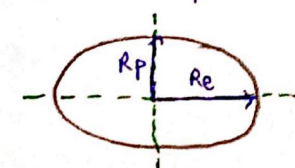
$$\omega^2 R = 0.034 \text{ rad/s}^2$$

Also,

$$\text{At equator} \Rightarrow g_{eff} = g - \omega^2 R \quad (\theta = 0^\circ)$$

$$\text{At poles} \Rightarrow g_{eff} = g \quad (\theta = 90^\circ)$$

→ Real shape of earth is not spherical but 'Ellipsoid', and there is diff. b/w radius at poles & equator.

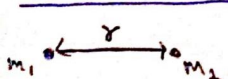


$$R_e - R_p \approx 20 \text{ to } 30 \text{ km}$$

$$g_p - g_e \approx 0.03 \text{ m/s}^2$$

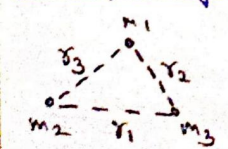
Gravitational Potential Energy:

For a two mass system



$$U = -G \frac{m_1 m_2}{r}$$

For a system of three masses



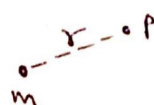
$$U = - \left[\frac{G m_1 m_2}{r_{12}} + \frac{G m_1 m_3}{r_{13}} + \frac{G m_2 m_3}{r_{23}} \right]$$

$$W_{by \text{ ext.}} = U_f - U_i$$

$$W_{by \text{ grav. Force/Field}} = U_p - U_f$$

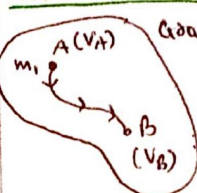
Gravitational Potential:

→ At a point, it is the potential energy of the given mass with a unit mass.



$$V = -\frac{GM}{r}$$

Work done in Grav. Field →



Grav. Field

Without even knowing displacement, we can find work done.

$$W_{ext.} = m(V_B - V_A)$$

$$W_{gf.} = m(V_A - V_B)$$

$$U = mV$$

Relation in $\vec{g}$ & V :

$\vec{F} = -\nabla U$, in any conservative force field.

$$\Rightarrow \vec{g} = -\nabla V$$

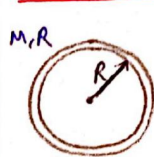
$$\Rightarrow \vec{g} = - \left[\frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k} \right]$$

(For 3D variation in V)

$$\vec{g} = -\frac{dV}{dr} \hat{r}$$

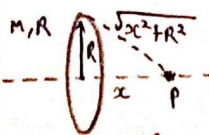
(For unidirectional variation in V)

Gravitational Potential due to Ring:

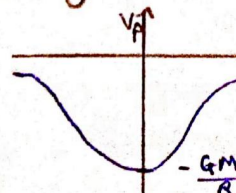


At centre of the Ring,

$$V_c = -\frac{GM}{R}$$



$$V_p = -\frac{GM}{\sqrt{R^2 + x^2}}$$



Also, the work done in lifting m to h height is

$$W_{ext.} = U_f - U_i = -\frac{GMm}{(R+h)} + mgR \Rightarrow$$

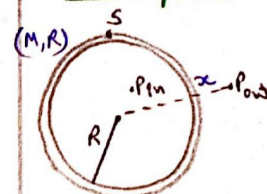
Gravitational potential due to sphere:

We can write grav. potential by

$$V = - \int_{\infty}^P \vec{g} \cdot d\vec{x}$$

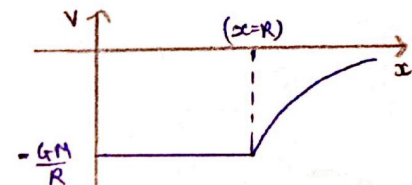
(P → Point at which we want V)

1) Hollow sphere →

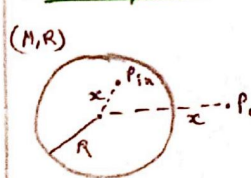


$$V_{out} = -\frac{GM}{x}$$

$$V_{in} = V_s = -\frac{GM}{R}$$



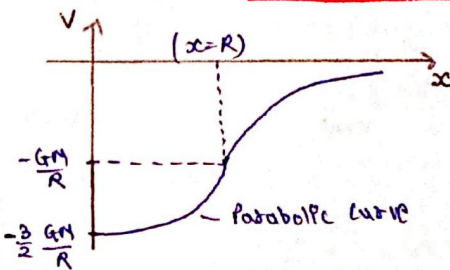
2) Solid sphere →



$$V_{out} = -\frac{GM}{x}$$

$$V_s = -\frac{GM}{R}$$

$$V_{in} = -\frac{GM}{2R^3} (3R^2 - x^2)$$



Self Energy:

Hollow (M, R)



$$E = -\frac{GM^2}{2R}$$

Solid (M, R)



$$E = -\frac{3GM^2}{5R}$$

Grav. Potential Energy of a body on Earth:

$$PE = mV = m \left(-\frac{GM_e}{R_e} \right) \Rightarrow U = -\frac{GM_e m}{R_e}$$

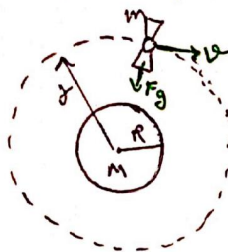


$$U = -mgR_e \quad \left(g = \frac{GM_e}{R_e^2} \right)$$

$$W = -\frac{GM_e m}{R_e^2} \cdot R_e + mgR_e \quad (\text{Only when } h \ll R_e) \Rightarrow W = mgh$$

Understanding of Satellite Motion:

Consider a satellite of mass m orbiting around a planet of mass M .



Orbital Velocity (v_o)

$$\frac{GMm}{r^2} = \frac{mv_o^2}{r} \Rightarrow v_o = \sqrt{\frac{GM}{r}}$$

Orbital Period (T)

$$T = \frac{2\pi r}{v_o} = \frac{2\pi}{\sqrt{GM}} (r)^{3/2}$$

$$\Rightarrow T \propto (r)^{3/2}$$

Interaction Energies

$$KE = \frac{1}{2}mv_o^2 = \frac{GMm}{2r} \quad PE = -\frac{GMm}{r}$$

$$TE = KE + PE = -\frac{GMm}{2r}$$

$$\text{So, } |TE| = |KE| = \frac{1}{2} |PE|$$

Escape Velocity:

For a body to escape earth's gravitational field, its total energy at instant of escape should be zero.

Diagram: A satellite of mass m is launched from the surface of Earth (radius R_e , mass M_e) with initial velocity v_e . At infinity, $v \rightarrow 0$, $KE = 0$, and $PE = 0$.

PE of mass m is $U = -\frac{GM_em}{R_e}$ (-ve sign denotes boundness)

To escape Earth's gravity.

$$TE = -\frac{GM_em}{R_e} + \frac{1}{2}mv_e^2 = 0$$

$$\Rightarrow v_e = \sqrt{\frac{2GM_e}{R_e}}$$

$$v_e = \sqrt{2gR_e} \quad \left[\text{For earth, } v_e = 11.2 \text{ km/s} \right]$$

Any object having $v > \sqrt{2gR_e}$ would leave Earth's gravity.

Gravitation-2

Escaping of a Satellite:

Diagram: A satellite is shown at a distance r from a planet of mass M_e with radius R_e . The orbital velocity is $v_o = \sqrt{\frac{GM}{r}}$. The escaping velocity is v_{SE} at $v = v_{SE}$.

Let the escaping velocity of satellite be v_{SE} , then at $v = v_{SE}$

$$TE = -\frac{GMm}{r} + \frac{1}{2}mv_{SE}^2 = 0$$

$$\Rightarrow v_{SE} = \sqrt{\frac{2GM}{r}}$$

$$\Rightarrow v_{SE} = \sqrt{2} v_o$$

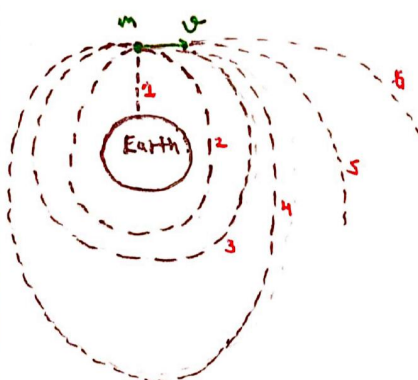
Increase in velocity

$$\Delta v = (\sqrt{2}-1)v_o$$

$$\Rightarrow \Delta v = 0.414 v_o$$

(41.4% of orbital velocity)

Projected Velocity vs Path:



If $v=0 \Rightarrow$ Path 1 (Straight Line)

$0 < v < v_o \Rightarrow$ Path 2 (Elliptical)

$v = v_o \Rightarrow$ Path 3 (Circular)

$v_o < v < v_e \Rightarrow$ Path 4 (Elliptical)

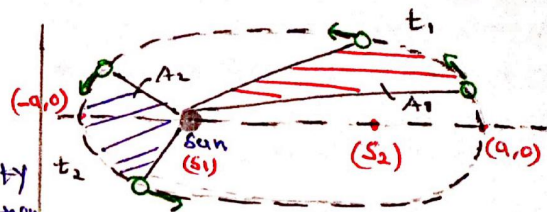
$v = v_e \Rightarrow$ Path 5 (Parabolic)

$v > v_e \Rightarrow$ Path 6 (Hyperbolic)

Kepler's Law:

1) Law of orbits $\Rightarrow$ Planet revolves around sun in an elliptical path with sun at one of its foci.

2) Law of Area $\Rightarrow$ Line joining sun and planet sweep equal area in equal time interval.



ie $\frac{dA}{dt} = \frac{L}{2m} = \text{const.}$ $\left[\frac{L}{2m} = \frac{\pi ab}{T} \right]$
($L \rightarrow$ Ang. momentum) $b \rightarrow$ Minor axis
 $a \rightarrow$ Major axis

Also,

$$\frac{A_1}{t_1} = \frac{A_2}{t_2}$$

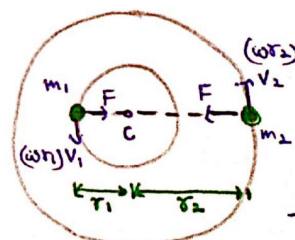
3) Law of Periods $\Rightarrow T^2 \propto a^3$

$T \rightarrow$ Period of oscillation of planet
 $a \rightarrow$ Semi-major axis of ellipse

$$\Rightarrow T^2 = \frac{4\pi^2}{GM_s} \cdot a^3$$

[$M_s \rightarrow$ Mass of Sun]

Binary Star System:



$\rightarrow$ Both mass rotate about their common centre of mass (C).

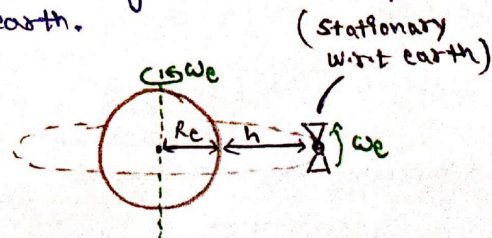
$\rightarrow$ Both m_1 & m_2 rotate with same ω & have same T .

$$\omega = \sqrt{\frac{G(m_1+m_2)}{r^3}} \quad (r = r_1 + r_2)$$

Also, $\frac{v_1}{v_2} = \frac{r_1}{r_2}$

Geostationary Satellite:

$\rightarrow$ A satellite which has same angular velocity as earth (ω_e) and appears stationary w.r.t its launch point on earth.



Conditions :

(i) $T = 24 \text{ hours}$

(ii) Must lie in equatorial plane of earth

(iii) Must revolve along the dirn of earth's rotation.

Calculations :

$$T^2 = \frac{4\pi^2}{GM_e} (R_e + h)^3$$

(From Kepler's 3rd law)

Since, $g = \frac{GM_e}{R_e^2}$

So, $T^2 = \frac{4\pi^2 R_e^2}{GM_e} \frac{(R_e + h)^3}{R_e^2}$

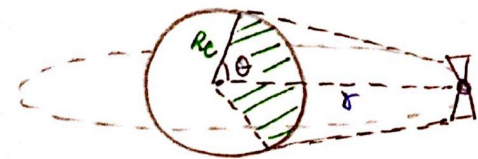
$$T^2 = \frac{4\pi^2}{g} \frac{(R_e + h)^3}{R_e^2}$$

$$\Rightarrow h = \left[\frac{T^2 R_e^2 g}{4\pi^2} \right]^{1/3} - R_e$$

For Earth,

$$h = 3.6 \times 10^4 \text{ km} \\ = 36000 \text{ km}$$

Broadcasting Region of a Satellite $\Rightarrow$



$$\theta = \cos^{-1} \left(\frac{R_e}{r} \right)$$

And Solid angle made by the satellite's cone broadcasting region,

$$\Omega = 2\pi (1 - \cos \theta)$$

$$\Rightarrow \Omega = 2\pi \left(1 - \frac{R_e}{r} \right)$$

So, broadcasting region is simply,

$$S = \Omega R_e^2$$